Zahra Khan

Location: Islamabad, Pakistan Email:getzahrakhan@gmail.com LinkedIn Portfolio Github

Profile

Robotics and AI Engineer with expertise in autonomous systems, robot design, motion planning, perception, and decision-making. Skilled in developing and evaluating algorithms for intelligent systems, including reinforcement learning, SLAM, and trajectory optimization. Experienced in integrating machine learning with robotics for adaptive and efficient behavior. Passionate about advancing research in robotic autonomy, swarm robotics, and space/field robotics applications.

Area of expertise include:

- **Robotics:** Robot Operating System (ROS1 & ROS2), SLAM, Motion Planning, Trajectory Optimization, Differential Drive Robots, Quadcopter Control, Mavlink/PX4, Surgical Robotics
- **Artificial Intelligence:** Reinforcement Learning, Deep Learning, Computer Vision, Neural Networks, Large Language Models (LLMs)
- **Programming:** Python, C++, MATLAB, Raspberry Pi, SITL/HITL Simulations
- **Simulation & Tools:** Gazebo, MuJoCo, Stable-Baselines3, Gymnasium
- **Software & OS:** Linux, Git, Agile
- **Languages:** English (Proficient)

Objective: To contribute to cutting-edge research in robotics and intelligent systems, focusing on autonomy, robot design, swarm systems, and algorithm development in real-world and field environments.

Work Experience

AI Researcher – Center of Excellence in Artificial Intelligence, Baharia University
Islamabad, Pakistan – October 2024- Present

- Contributed to industrial projects, including AI-driven drone technologies for warehouse automation.
- Applied Large Language Models (LLMs) for automated literature review and adaptive decision-making systems.
- Designed AI-based solutions for industrial applications.
- Designed AI-based systems for clinical diagnostics and robotic workflows.
- Collaborated with teams to ensure timely project delivery and knowledge sharing.

Research Student – Intelligent Robotics Lab (NCAI), National University of Science and Technology (NUST)
Islamabad, Pakistan – 2022- 2025

- Conducting advanced research on motion planning in cluttered environments using reinforcement learning techniques.
- Developed and implemented novel algorithms to optimize path planning in complex environments, contributing to advancements in autonomous navigation.
- Collaborating with a multidisciplinary team to integrate reinforcement learning-based models into autonomous robotic systems for real-world applications.

Teaching Assistant - National Vocational and Technical Training Commission (NAVTTC)
Islamabad, Pakistan – May 2024 – July 2024

- Delivered courses on Robotics, Artificial Intelligence, 3D Printing, Arduino Programming, and Machine Learning to students, focusing on both theoretical and practical aspects.
- Developed comprehensive instructional materials and provided hands-on guidance to ensure effective learning.
- Fostered an engaging and interactive learning environment, encouraging student participation and enhancing their problem-solving skills.

Research Intern - Aerial Robotics Lab National University of Science and Technology (NUST)
Islamabad, Pakistan - June 2022 – Aug 2022

- Spearheaded research involving the development of a Machine Learning model for the detection of COVID-19 in HRCT lung images.
- Curated and processed HRCT images of COVID-19 patients, preparing them for analysis.
- Employed Transfer Learning techniques to create a robust model achieving 86% accuracy in identifying COVID-19 from HRCT images.

Research Intern - International Center of Chemical and Biological Sciences (ICCBS)
Karachi, Pakistan - June 2021 – Aug 2021

- Ensured safety and functionality of various spectrometry and Electrical equipment through effective maintenance procedures.
- Conducted field surveys and analyzed data to troubleshoot and resolve power systems and equipment issues.
- Supervised the installation of electrical systems within the building, ensuring adherence to industry standards.

Relevant Projects

Autonomous UAV with SLAM-based Navigation – Bahria University, Islamabad

- Assembled and programmed a Holybro X500 V2 quadrotor for autonomous flight.
- Implemented **SLAM-based localization**, **obstacle avoidance**, and **sensor fusion** for robust navigation.
- Tested autonomous flight in complex environments, considering **sensor power and payload constraints**.

Automated Literature Review with Large Language Models – Center of Excellence in AI, Islamabad, Pakistan

- Developed a Python/Flask system using LLaMA 3.0 to extract, summarize, and synthesize robotics & AI research.
- Built a web interface for efficient querying and knowledge discovery.
- Integrated RL concepts with LLMs for adaptive research automation.

Robot Trajectory Planning – NUST, Islamabad, Pakistan

- Researched and implemented trajectory optimization algorithms in MATLAB.
- Enhanced robotic motion efficiency in simulated environments.

Motion Planning with SLAM Integration - IRL, NCAI, NUST, Islamabad, Pakistan

- Integrated SLAM-based localization with motion planning algorithms.
- Compared performance of DWA, TEB Planner, and MPC Planner.

COVID-19 Detection in HRCT Scans - Aerial Robotics Lab, NUST, Islamabad, Pakistan

- Developed a robust machine learning model for COVID-19 diagnosis with 95% accuracy.
- Leveraged advanced image processing and transfer learning techniques.

Image Classification with UNET and DeepLabv+ - NUST, Islamabad, Pakistan

- Implemented state-of-the-art image segmentation models for classification tasks.

Education

MS in Robotics and Intelligent Machine Engineering - National University of Science and Technology (NUST), Islamabad, Pakistan – 2021 - 2025
CGPA: 3.80/4.00 | Thesis: Motion planning of Differential Derive robots in Cluttered Environments using Deep Reinforcement Learning
BS in Electrical Telecommunication Engineering – COMSATS University Islamabad, Pakistan – 2016 - 2020
CGPA: 3.12/4.00 | Thesis: designing of the holographic microscopic device for the detection of pathogenic samples.

Certifications

- **Da Vinci Surgical Robot Workshop 1** – Johns Hopkins University
- **Da Vinci Surgical Robot Workshop 2** – Johns Hopkins University:
  Hands-on training in robotic-assisted surgery using Da Vinci system, emphasizing surgeon-robot interaction, control systems, and clinical integration.
- **Workshop on Deep Reinforcement Learning** – Coventry University
- **AI- NUEX Tutor** – National Center of Artificial Intelligence
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Honors and Awards

- **Prime Minister's Laptop Scheme**: by Higher Education Commission (HEC) for academic excellence.
- **Leader Shift Program**: Selected for leadership initiatives at COMSATS University.

Publications

Z. Khan, K.F. Iqbal, Y. Ayaz, and M.T. Nasir, "A Map-Informed Dense Reward Function for Efficient Robot Motion Planning Using Deep Reinforcement Learning", Knowledge-Based Systems **(Submitted)**
Impact Factor: 7.8

References

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